

Bachelor Thesis

Nsak as Framework for Scenario Based Network Security

Course of study	Bachelor of Science in Computer Science
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Abstract

One-paragraph summary of the entire study – typically no more than 250 words in length (and in many cases it is well shorter than that), the Abstract provides an overview of the study.

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1. Introduction

Computer networks are constantly exposed to evolving cyber threats ...

The *Network Swiss Army Knife (NSAK)* is a framework designed to execute orchestrated security scenarios consisting of reusable attack drills in a controlled network environment. The framework aims to support automated security testing and adversary emulation on network edge devices.

The development of the NSAK framework originates from previous project work, in which the core architecture and the basic scenario execution model were developed. This thesis builds upon this foundation and extends the framework by investigating how AI agents can support automated red and blue team scenarios.

...

1.1. Motivation

Hypothesis: AI can enhance red and blue team simulations in a network environment by generating and executing scenarios within the NSAK framework and assisting in the evaluation of detection results.

1.2. Goals

For a goal to be considered **S.M.A.R.T.**, it must fulfill the following criteria:

- ▶ **Specific:** What do we want to achieve?
- ▶ **Measurable:** What are the success criteria?
- ▶ **Achievable:** How do we plan to achieve this goal?
- ▶ **Relevant:** Why is it important to achieve this goal?
- ▶ **Time-bound:** Goals categorized as *must* have to be achieved within the thesis, while *should* or *could* goals are considered optional or nice to have.

1.2.1. Research

Compare Frameworks (must)

To identify the USP (unique selling point) of the NSAK framework, we will analyze and compare existing security emulation frameworks. First, relevant frameworks such as MITRE Caldera and Atomic Red Team will be identified and reviewed. Based on this analysis, the identified concepts will be compared with the architecture of the NSAK framework. The comparison will focus on how scenarios and drills are defined, how configuration and parameter handling are structured, how reusable components are organized, how automation is achieved, and which AI capabilities they provide.

The analysis and comparison will be based on the following properties:

- ▶ Concepts
- ▶ Modularity
- ▶ Configurability
- ▶ Automation
- ▶ AI Capabilities

Success criteria:

- ▶ At least two security emulation frameworks are identified and reviewed
- ▶ A matrix comparing the specified properties of the selected frameworks is created
- ▶ The USP of the NSAK framework is identified

AI Integration (must)

The goal of this research is to investigate how AI (artificial intelligence) can support automated security testing and adversary emulation. The research evaluates potential integration points where AI could enhance the scenario execution, automation, decision-making, and analysis capabilities of the NSAK framework.

Success criteria

- ▶ How is AI currently used in cybersecurity automation?
- ▶ Which tasks in adversary emulation can be supported or automated by AI?

- ▶ How could AI interact with the NSAK scenarios and drills? (e.g. Agents, MCP)
- ▶ What architectural changes would be required to integrate AI into NSAK?

1.2.2. Framework

REST API (TBD)

Adding a REST API to the NSAK framework was already proposed in the predecessor project, as it greatly enhances interoperability with other tools. The priority of this goal depends heavily on the outcome of the goal “Research: AI Integration” and on whether we want to fulfill the goal “Framework: Web GUI”, which is prioritized as “could”. If the evaluated AI integration requires a REST API (e.g., MCP), this goal needs to be prioritized as “must”; it will be prioritized as “could”.

Success criteria:

- ▶ A REST API specification with feature parity to the CLI is created.
- ▶ The REST API is implemented according to the specification.
- ▶ The API is documented (Open API) and usable by external clients.
- ▶ The REST API remains optional and does not replace the CLI-based interaction.
- ▶ Authentication and Authorization are explicitly out of scope.

MCP Server(TBD)

The MCP Protocol allows the Ai to interact with applications in a standardized way - and could therefore be necessary for implementing AI-based scenarios. If the outcome of the research task ”AI Integration” highlights the necessity to specify/expose an MCP-Server to interact seamlessly with the NSAK framework, the implementation is a (must), otherwise a (could).

- ▶ A MCP-Server is specified and implemented
- ▶ Documentation how to connect to the MCP is provided.
- ▶ A form of authentication is provided.

Web GUI (could)

The goal is to design and implement a web-based graphical user interface that allows users to interact with the NSAK framework in a user-friendly way. The Web GUI will enable management of scenarios and drills, triggering scenario execution, and visualization of execution results and system status.

Success criteria:

- ▶ A simple Web GUI that interacts with the REST API has been implemented.
- ▶ The Web GUI displays a list of available scenarios and their states.
- ▶ The scenario's results can be viewed on the Web GUI
- ▶ Scenarios can be configured and executed via the Web GUI
- ▶ Authentication and authorization are explicitly out of scope.

1.2.3. Scenarios

AI – Purple Team (must)

The goal is to implement artificial intelligence according to the result of “Research: AI Integration” and to explore the use of AI to support red, blue, and purple team exercises in the NSAK framework. The AI will be evaluated for its potential to assist in executing attacks in a scenario and analyzing detection results. The findings will help assess how AI could enhance and coordinate red and blue team activities across a network via the NSAK device.

Success criteria:

- ▶ The AI is integrated into NSAK as a scenario or as part of the framework, depending on the result of “Research: AI Integration”.
- ▶ The AI can execute CLI or Python tools to directly or indirectly interact with the attached network.
- ▶ An operator can interact with or stop the running AI.
- ▶ The result artifacts are stored and can be presented or reused for further scenarios.

Reproducible Test Environment (must)

The goal is to design and implement a reproducible and safe test environment for the NSAK framework that enables consistent execution and evaluation of scenarios and drills. The environment should provide a stable basis for testing and comparison of manual and AI-assisted scenarios.

Success criteria:

- ▶ The test environment can be set up reproducibly using a defined configuration or setup procedure with Infrastructure as code (e.g., Ansible)
- ▶ NSAK scenarios and drills can be executed consistently within the environment.
- ▶ The environment supports both manual and AI-assisted scenarios.
- ▶ Scenario results can be collected and analyzed for evaluation purposes.

Intrusion Detection – Blue Team (should)

The goal is to define and implement a blue team reference scenario for intrusion detection within the NSAK framework. The scenario should allow observation and evaluation of suspicious network activity to assess detection capabilities in the test environment and compare the results with those of the AI approach.

Success criteria:

- ▶ A blue-team intrusion-detection scenario is implemented in the NSAK framework.
- ▶ Detection events or alerts can be observed and documented during scenario execution.
- ▶ Detection results can be compared with results from AI-assisted scenario execution.

Network Discovery – Red Team (must)

The goal is to implement a reference red team scenario for network discovery within the NSAK framework. The scenario should simulate reconnaissance activities to identify hosts, services, and network characteristics, and the results should be comparable to the AI-Agent approach.

Success criteria:

- ▶ A red team network discovery scenario is implemented in the NSAK framework.
- ▶ Discovered network information is collected as artifacts.
- ▶ The scenario can be executed reproducibly within the test environment.
- ▶ The results can be compared with those generated by the AI-assisted approach.

1.2.4. Evaluation Goals

AI Scenario (must)

The objective is to determine the effectiveness of the AI-based scenario introduced by the goal “AI – Purple Team (must)” within the network environment defined by the goal “Reproducible Test Environment (must)”. The evaluation will assess whether the AI scenario successfully executed the expected red and blue team activities for the provided prompts and the quality of the results.

Success criteria:

- ▶ A methodical approach for the assessment is evaluated and documented.
- ▶ The results of the AI scenario for red and blue team activity are assessed.
- ▶ There is a conclusion on the effectiveness of the AI scenario for red and blue team activities.

AI vs Engineered Scenarios (must/should)

The goal of this evaluation is to compare the quality of AI-based red (network reconnaissance) (must) and blue (network intrusion detection) (should) team scenarios with that of manually engineered scenarios.

The comparison focuses on criteria such as execution reliability, reproducibility, and coverage. The results will help determine the potential benefits and limitations of AI-supported scenarios and provide insights into how AI could enhance future versions of the NSAK framework.

Success criteria:

- ▶ A methodical approach for the comparison is evaluated and documented.
- ▶ The results of the AI scenario are compared against the respective results of the red team scenario (must)
- ▶ The results of the AI scenario are compared against the respective results of the blue team scenario (should)
- ▶ There is a conclusion on the effectiveness of the AI scenario compared to manually engineered scenarios.

2. Related Research

2.1. Comparison Nsak - Caldera

3. Concepts - Framework

4. Methodic

5. Implementation

6. Evaluation

7. Discussion

8. Conclusion

List of Figures

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Glossary

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A. Project Management

A.1. Agile Management Process

The project followed an interactive development approach using Gitlab for issue tracking and planning. Work items were organized into epics, milestones and issues. Each iteration lasted two weeks.

A.1.1. Issue Lifecycle

The workflow used for ticket tracking consisted of the following stages

- ▶ Backlog
- ▶ Todo
- ▶ Q&A
- ▶ Done

A.2. Definition of Ready (DoR)

An ticket is considered ready for implementation when the following criteria are fulfilled.

A.2.1. Clarity

- ▶ Clear and precise title
- ▶ Objective or problem described in two to three sentences
- ▶ Scope clearly defined

A.2.2. Scope

- ▶ Affected module specified (drill, scenario, core, container, frontend)
- ▶ External dependencies identified

A.2.3. Validation

- ▶ Testable acceptance criteria defined
- ▶ Expected behavior clearly specified

A.3. Definition of Done (DoD)

A ticket is considered done when all the following criteria are fulfilled.

A.3.1. Implementation

- ▶ All acceptance criteria fulfilled
- ▶ Container execution verified
- ▶ Pre-commit hooks pass
- ▶ Logging and error handling consistent

A.3.2. Testing

- ▶ Feature or bug fix tested
- ▶ Acceptance criteria verifiable (automated or manual)
- ▶ Relevant edge cases considered
- ▶ Execution reproducible

A.3.3. Documentation

- ▶ README updated if required
- ▶ Architecture changes documented
- ▶ Thesis relevance briefly noted

- ▶ Limitations documented
- ▶ Required documentation stored for thesis or appendix

A.4. Project Structure

A.4.1. Milestones

Milestones represent crucial steps toward the overall project goal.

A.4.2. Epics

Epics consist of multiple stories and contribute to fulfilling milestones.

A.4.3. Stories

Each story is assigned a priority, sprint, milestone and epic. Stories must meet the Definition of Ready before implementation.

A.5. Planning and Estimation

The following estimation strategy was used:

- ▶ 1 hour corresponds to 1 weight point
- ▶ Iteration length: 2 weeks
- ▶ Sprint Planning duration: 1.5 hours per iteration
- ▶ Sprint Review duration: 1 h per iteration
- ▶ Sprint Retro duration: 1 h per iteration